Kirolos Sedra

Data Analyst
Waterloo, Ontario | Relocatable | Authorized to work in Canada; no sponsorship required
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Professional Summary

Data Analyst with hands-on experience using **Python**, **SQL**, **Power BI**, and **Excel** to clean, analyze, visualize, and interpret experimental, IoT, and fintech data. Experienced in EDA, statistical analysis, KPI reporting, dashboarding, and translating empirical evidence into engineering and stakeholder decisions through industry-collaborative research, with a **Pattern Analysis and Machine Intelligence** specialization.

Experience

Wireless Sensors and Devices Lab, University of Waterloo **Sep 2024 – Present**
Wireless Data Analyst (Research Assistant) *Full-time, On-site*

- Designed and analyzed **80 controlled experimental runs** across four scenarios (20 one-minute runs per scenario), building **Python-based data pipelines** to collect, clean, merge, align, filter, and aggregate high-frequency wireless and performance datasets.
- Performed **EDA and comparative analysis** across mobility, load, interference, line-of-sight, and attenuation conditions using counts/percentages, heatmaps, time-series charts, box plots, cross-metric overlays, and frequency–time visualizations; investigated anomalous behavior using synchronized logs and experimental context.
- Applied distribution-aware statistics using **P50/P99**, mean/standard deviation, one-sample left-tailed **t-tests**, p-values, threshold-based degradation criteria, and **Holm correction** across repeated trials to distinguish systematic performance changes from run-to-run variation.
- Supported **Apple engineering decisions through empirical analysis**, translating controlled experimental results into weekly stakeholder presentations, visual evidence, and technical recommendations; communicated findings to industry partners, Senior Vice Presidents, and mixed technical/non-technical audiences.
- Combined timestamped network, performance, and GPS datasets to build street-level heatmaps and explain location-, mobility-, and network-driven changes, turning multiple raw data sources into decision-ready findings.

Aquasensing **Sep 2024 – Present**
Firmware Specialist / Research Intern *Part-time, Startup Environment*

- Integrated **Power BI dashboards into a React application** for real-time IoT monitoring and operational visibility, connecting device-facing workflows with stakeholder-ready analytics for customer demonstrations and ongoing system review.
- Analyzed device telemetry, logs, RF measurements, and end-to-end test results to diagnose production issues; built **Python–SQL** tooling and used stored-procedure logging and rollback safeguards to improve data integrity and reproducibility.
- Translated technical measurements and operational findings into stakeholder-facing conclusions for **12Energy**, **Evergreen**, and **Bell IoT**, contributing to deployment planning and a **50-gateway commercial rollout**.

ChurchOne / LinC **Ongoing**
Product Owner & Solution Architect *Volunteer, Digital Transformation Initiative*

- Led a five-person initiative and architected a **React**, **TypeScript**, and **Firebase** platform used by **30+ congregation members**, defining structured data workflows, validation rules, and reporting views for attendance, people records, assignments, forms/surveys, and notes so records could be consistently captured, searched, reviewed, and used for follow-up and decision-making.

Ejada Systems **Sep 2023 – Aug 2024**
Software Engineer *Full-time, Hybrid*

- Developed fintech REST APIs and backend services using **Java Spring Boot**, **SQL**, and stored procedures for **Al Rajhi Bank** and **Emkan Governmental Services**.
- Improved database-query performance by **30%** through SQL optimization, indexing, and stored-procedure redesign; improved API performance by **50%** by eliminating redundant database/service calls and optimizing retrieval logic.
- Performed database-side debugging, data validation, and stored-procedure analysis while maintaining operation logging and rollback scripts for safe, traceable schema and data changes.

Selected Data & Machine Learning Coursework

ECE 657A – Data Analysis and Machine Learning *Python, pandas, scikit-learn, TensorFlow/Keras*

- Completed EDA/modeling assignments on Wine Quality, Abalone, and Forest Fires datasets covering missing-data assessment, descriptive statistics, distributions, class balance, standardization, train/test splits, **5-fold cross-validation**, and classification/regression evaluation.
- Implemented KNN, Decision Trees, Random Forests, Gradient Boosting, and hyperparameter searches; analyzed CNN representations on Fashion-MNIST using **PCA**, explained variance, **t-SNE**, and **K-Means** clustering.

Additional Applied Analytics & ML Coursework *Computational Intelligence, Image Processing, NLP*

- Applied normalization, standardization, missing-value imputation, one-hot encoding, confusion matrices, ROC/AUC, and train/validation/test methodology; implemented neural-network training concepts including backpropagation, Adam, validation tracking, and early stopping.
- Built text-analysis workflows using tokenization, unigrams/bigrams, CountVectorizer, and Multinomial Naive Bayes; also worked with Word2Vec, PCA/ICA concepts, image/video analysis, and model/runtime trade-off evaluation.

Education

University of Waterloo **Sep 2024 – Aug 2026**
Master of Applied Science in Electrical and Computer Engineering *92% (A+)*
Specialization: Pattern Analysis and Machine Intelligence

Ain Shams University **Sep 2018 – Jul 2023**
Bachelor of Science in Computer Engineering *CGPA: 3.12/4.00*

Technical Skills

Data & Programming: Python, SQL, pandas, NumPy, scikit-learn, MATLAB, Java, Bash
BI & Visualization: Power BI, Excel (charts, dashboards, statistical analysis, VBA/macros), Tableau (coursework), heatmaps, time-series charts, box plots, publication-quality figures/tables
Analysis & Statistics: EDA, data cleaning, descriptive statistics, P50/P99, mean/standard deviation, hypothesis testing, t-tests, p-values, Holm correction, cross-validation
Machine Learning: Regression, classification, KNN, Decision Trees, Random Forests, Gradient Boosting, Naive Bayes, K-Means, PCA, t-SNE, CNNs, TensorFlow/Keras
Data Systems: Relational databases, stored procedures, indexing, REST APIs, Azure IoT, logging and rollback/data-integrity workflows
Spoken Languages: English (C1), French (B1)

Leadership & Communication

- University of Waterloo Teaching Assistant for ECE 106, ECE 610, and ECE 358; IEEE AP-S Chapter Webmaster and Treasurer; experienced presenting analytical results, evidence-based recommendations, and technical trade-offs to industry, executive, and non-technical audiences.